

Due date: Wednesday 8:50am, Sep. 30, 2026

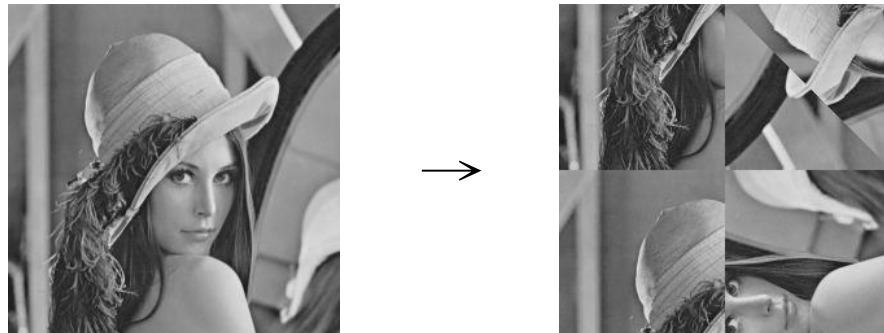
1 Raw image I/O (70%)

1.1 Raw image file format understanding.

- Download raw images from ntut course system. <http://istudy.ntut.edu.tw/learn/>
- Find a raw image viewer that you prefer to read raw image *lena256.raw*. One of the choices is to use Xnview from <http://www.xnview.com>. Save and show the screenshot of output. What options do you have to set to read the raw images correctly? (Figure, 5%; Discussion, 5%)

1.2 Raw image file input/output. (OpenCV is not allowed in this section)

- Download and modify the example program, IOTest.zip, to read the raw image file *lena256.raw*.
 - Let the origin of pixel coordinate be the 0th row & 0th column located at the top left corner pixel of images (to follow the C/C++ convention). What is the pixel coordinate (row, column) of the pixel whose intensity value is 232 in *lena256.raw* file(only print first one)? (10%)
 - What is the pixel order (the nth pixel, start counting from 0) of the 20th occurrence of the pixel with intensity value 150 in *lena.256.raw* file? (10%)
- Read the image *lena256.raw* and divide into 2 triangular sub-blocks and 3 square sub blocks, and then flip or rotate and exchange the sub-blocks. Write a C/C++ program to modify the original image and save as *lena256_out.raw*. The result should be the same as shown below. Save and display the result. (Figure, 10%; Discussion, 10%)



1.3 Brightness

- Reading *lena256.raw* and increase the image brightness with random value between -60 to 60. Write the output to a raw image file. Use Xnview to view the result. (Figure, 5%)
- What should you have to be careful of in part a? (Hint: overflow/underflow with data type) (Discussion, 5%)

2 OpenCV image I/O (C/C++ with OpenCV) (30%)

This problem is to practice calling OpenCV functions. You may learn and use Drawing API from the following link: <http://docs.opencv.org/4.10.0/index.html>.

- Review and be familiar with C/C++ programming language and setup the OpenCV environment according to installation guide “[VS2022_OpenCV_SetupGuide.pdf](#)”
- Read *Fire.jpg* and try your best to find wildfire in image using OpenCV, and discuss what restriction you encounter while you are using OpenCV. Final wildfire results should be **bounded by bounding boxes** on the original image, as shown in the example below. Note: Any open-source AI technique is **prohibited**. Save the image as PNG format and use Xnview to verify your result. (Figure, 10%; Discussion, 20%)



example



Fire.jpg